

Booleans: True/False Conditions

Boolean values (or bool) represent true or false conditions. They're critical for decision-making in algorithms:

- Is the market open?
- Is this stock price above its moving average?
- Should we execute this trade?

Creating boolean variables:

```
1. is_market_open = True
2. is_price_above_average = False
3. should_buy = True
```

Converting Between Data Types

Sometimes you need to convert data from one type to another. Python makes this easy with built-in functions:

```
1. # Converting string to integer
2. volume_str = "15750000"
3. volume_int = int(volume_str)
4.
5. # Converting string to float
6. price_str = "142.58"
7. price_float = float(price_str)
8.
9. # Converting number to string
10. price = 142.58
11. price_str = str(price)
```

This conversion capability is particularly useful when processing financial data from various sources, like APIs or CSV files.

Operators: The Tools for Financial Calculations

Now that we can store data in variables, we need ways to manipulate this data. Operators are the tools that allow us to perform calculations, compare